A

Mini Project Report

on

Know Your Bite – (KYB)

Submitted in partial fulfillment of the requirements for the

degree

Second Year Engineering – Computer Science Engineering (Data Science)

by

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CERTIFICATE

This to certify that the Mini Project report on Know Your Bite – (KYB) has been submitted by Shivam Mewada (24107064), Nishant Navale (24107106) and Ashish Marothia (24107111) who are bonafide students of A. P. Shah Institute of Technology, Thane as a partial fulfillment of the requirement for the degree in **Computer Science Engineering (Data Science)**, during the academic year **2025-2026** in the satisfactory manner as per the curriculum laid down by University of Mumbai.

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Chapter 1 : Introduction

The project “Know Your Bite (KYB)” is a smart food analysis application designed to help users make healthier dietary choices. It allows users to scan barcodes, search products, or input ingredients manually to receive detailed insights. The system focuses on ingredient transparency and personalized health recommendations. By integrating multiple food databases, the app provides reliable and meaningful nutritional analysis. The goal is to promote awareness and improve everyday eating habits.

1.1. Purpose

The purpose of this document is to provide a detailed technical overview and implementation roadmap for the "Food Scanner" mobile application. The primary target audience includes project supervisors, academic evaluators, and mobile app developers interested in health-tech integrations. This report outlines how the system bridges the gap between complex chemical nomenclature on food labels and consumer understanding.

1.2. Problem Statement

Modern packaged food products often contain a long list of additives, preservatives, and stabilizers labeled with complex chemical names or E-numbers that the average consumer does not recognize. For individuals with chronic health conditions (like diabetes or hypertension) or severe allergies (gluten, nuts, dairy), these labels pose a significant health risk. There is a lack of an accessible, real-time tool that "translates" these labels into simple, actionable health insights.

1.3. Objectives

- To develop a user-friendly mobile application for scanning and analyzing food products using Flutter and REST APIs.
- To provide clear, non-technical explanations of ingredients using a curated database.
- To implement a real-time allergy detection system that alerts users to specific triggers.
- To promote healthy eating habits by generating a simplified "Health Score" based on nutritional composition.

1.4. Scope

The scope of Food Scanner extends to health-conscious consumers, fitness enthusiasts, and individuals with dietary restrictions. The system utilizes external Food Product APIs (like Open Food Facts) to fetch real-time data and matches it against an internal database of ingredient meanings. Future iterations could include AI-driven recommendations for healthier alternatives based on the user's specific health profile.

Chapter 2 : Proposed System

The proposed system is a mobile application that analyzes food products using barcode scanning, OCR, and search functionality. It retrieves product data from external sources and processes it through a rule-based engine. The system maps ingredients to a unified database to identify additives, preservatives, and their effects. It then evaluates nutritional values based on health standards and user conditions. Finally, it presents personalized insights in a structured and user-friendly format.

2.1. Features and Functionality

This section will describe the key features and core functionalities of the Know Your Bite (KYB) application. It will explain how the system will operate to deliver real-time food analysis and health insights to users. The features will be designed to ensure accuracy, usability, and efficiency by integrating technologies such as barcode scanning, API-based data retrieval, and intelligent data processing. Overall, this section will highlight how the application will translate complex product information into simple, actionable outputs that will support healthier decision-making.

- **Product Search/Scan:** Users can utilize the mobile camera to scan barcodes or manually search for products to retrieve data via API.
- **Ingredient Explanation:** The system breaks down complex names (e.g., "Maltodextrin") into simple terms, explaining its purpose and health impact.
- **Allergy Safety Alerts:** Immediate visual warnings if common allergens (nuts, soy, gluten) are detected.
- **Health Score Analysis:** An algorithmic evaluation of the product's nutritional value (sugar, fats, sodium) to provide a 1–10 health rating.
- **Clean Mobile Interface:** A responsive UI built with Flutter to ensure cross-platform compatibility and smooth user experience.

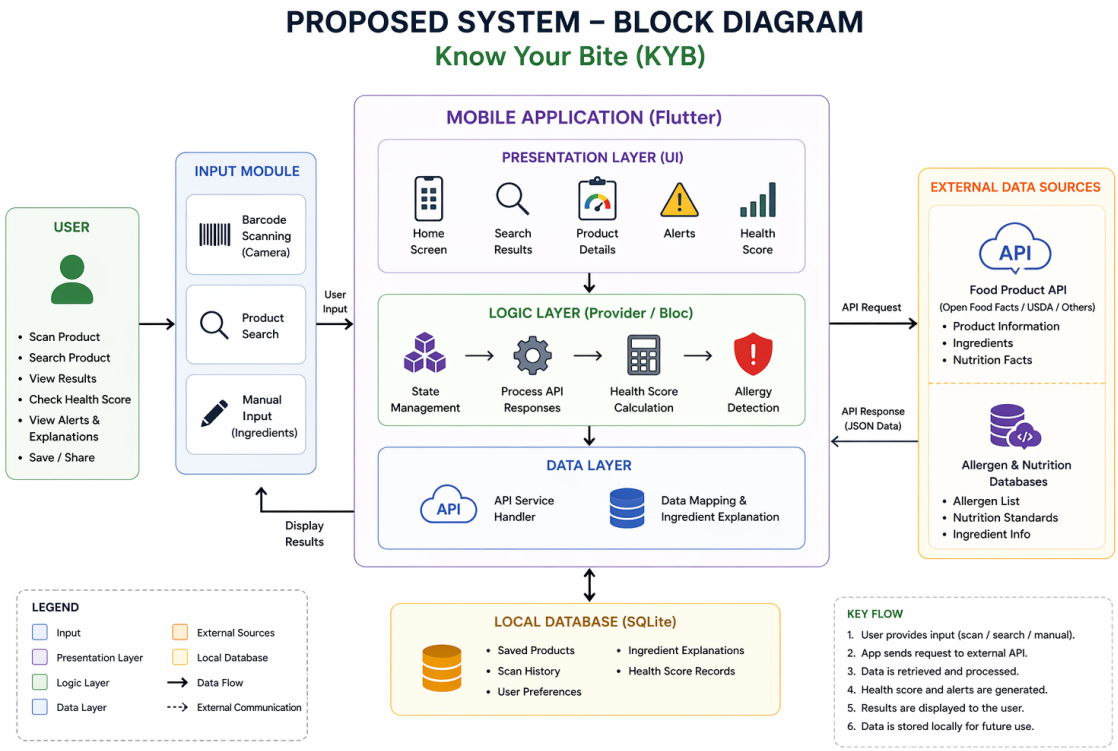

Figure 1 - Block diagram of Propsed solution.

The proposed system shown in **Figure 1 - Block diagram of Propsed solution** architecture of the Know Your Bite (KYB) application is designed to provide a structured and efficient flow of data between different components. It follows a modular approach, where the system is divided into input, processing, and data management units. This design ensures scalability, maintainability, and smooth interaction between the user and the application.

The workflow begins with the user, who interacts with the system through the mobile application. The user can scan a product barcode, search for a product, or manually enter ingredient details. These inputs are handled by the Input Module, which acts as the entry point of the system. It captures user data and forwards it to the main application for further processing.

The core of the system is the Mobile Application (Flutter), which is divided into three layers. The Presentation Layer manages the user interface, including screens such as home, search results, product details, alerts, and health score display. This layer ensures that all information is presented in a clear and user-friendly manner, allowing users to easily understand the results.

The Logic Layer is responsible for handling the internal processing of the system. It manages application state using tools like Provider or Bloc, processes API responses, calculates the health score based on nutritional values, and detects allergens using predefined rules. This layer acts as the brain of the system, converting raw data into meaningful insights.

The Data Layer handles communication with external and internal data sources. It sends API requests to external food databases such as Open Food Facts or USDA and receives product information in JSON format. Additionally, it maps ingredient data to an internal database that contains simplified explanations and nutrition rules. This layer ensures that accurate and relevant data is available for processing.

External data sources play a crucial role in the system by providing real-time product information, including ingredients and nutritional facts. The application communicates with these sources through APIs, ensuring that the data remains updated and reliable. Alongside this, a Local Database (SQLite) is used to store user-related data such as scanned products, history, preferences, and computed health scores for faster access and offline support.

Overall, the system follows a clear flow: user input is captured, processed through different layers, data is fetched and analyzed, and results are displayed back to the user. This layered and modular architecture enhances performance, simplifies development, and ensures that the KYB application delivers accurate, fast, and user-friendly health insights.

Chapter 3 : Project Outcome

The project successfully delivers a functional prototype capable of analyzing food products and generating health insights. Users can identify harmful ingredients and understand their effects clearly. The system provides personalized recommendations based on individual health conditions. It enhances awareness about food consumption and encourages healthier choices. Overall, the application demonstrates a practical approach to digital health assistance. successful implementation of the **FoodScanner** application aims to achieve the following tangible results:

- **Enhanced Consumer Awareness:** Users will transition from being passive consumers to informed decision-makers by understanding the "why" behind ingredient labels.
- **Immediate Health Risk Mitigation:** Through the automated allergy alert system, users can avoid medical emergencies related to accidental ingestion of allergens.
- **Nutritional Literacy:** The application simplifies complex data into a "Health Score," making it easier for users to compare two similar products and choose the healthier option.
- **Unified Food Database:** By integrating REST APIs with a custom ingredient meaning database, the project creates a comprehensive knowledge base for packaged goods.

Chapter 4 : Software Requirements

The system requires Flutter for frontend mobile app development and Dart as the programming language. It utilizes APIs such as Open Food Facts for product data retrieval. A local or cloud-based database is used to store ingredient information and user profiles. Development tools include Android Studio and necessary Flutter SDKs. The system also requires a device with camera support for scanning features.

4.1. Hardware Requirements

- **Processor:** Minimum Quad-core 2.0 GHz (for development)
- **RAM:** 8 GB Minimum (16 GB recommended for Android Studio/VS Code)
- **Device:** Android or iOS smartphone with a functional camera (minimum 8MP) for barcode scanning.

4.2. Software Requirements

- **Operating System:** Windows 10/11, macOS, or Linux.
- **Frontend Framework:** Flutter SDK (Dart Language).
- **IDE:** Visual Studio Code or Android Studio.
- **Database:** Firebase (for user data) and a local SQLite or JSON-based database for ingredient meanings.
- **APIs:** Open Food Facts API or similar RESTful services for product fetching.

Chapter 5 : Project Design

The project follows a modular architecture with separate layers for data retrieval, processing, and presentation. The frontend handles user interaction, scanning, and display of results. The backend logic processes ingredient data using a rule-based analysis engine. A unified ingredient database supports mapping and interpretation of complex food components. The design ensures scalability, maintainability, and efficient data flow between components.

5.1. System Architecture

This section will describe the overall system architecture of the Know Your Bite (KYB) application. It will explain how the system will be structured into different layers to ensure efficient data flow, scalability, and maintainability. The architecture will follow a layered approach, where each layer will have a specific responsibility, including user interaction, data processing, and data management. This structured design will help in separating concerns and improving the performance and reliability of the application.

1. **Presentation Layer (Flutter):** Handles the UI components, camera integration for scanning, and user input.
2. **Logic Layer (Provider/Bloc):** Manages the state of the application, processes API responses, and performs the health score calculation.
3. **Data Layer (REST API & Local DB):** Fetches external product data and maps ingredient strings to the internal explanation database.

5.2. User Interface Design

Each screen is designed for high readability. The "Result Screen" uses a color-coded system:

- **Green:** Safe/Healthy ingredients.
- **Yellow:** Processed/Cautionary ingredients.
- **Red:** Harmful additives or detected allergens.

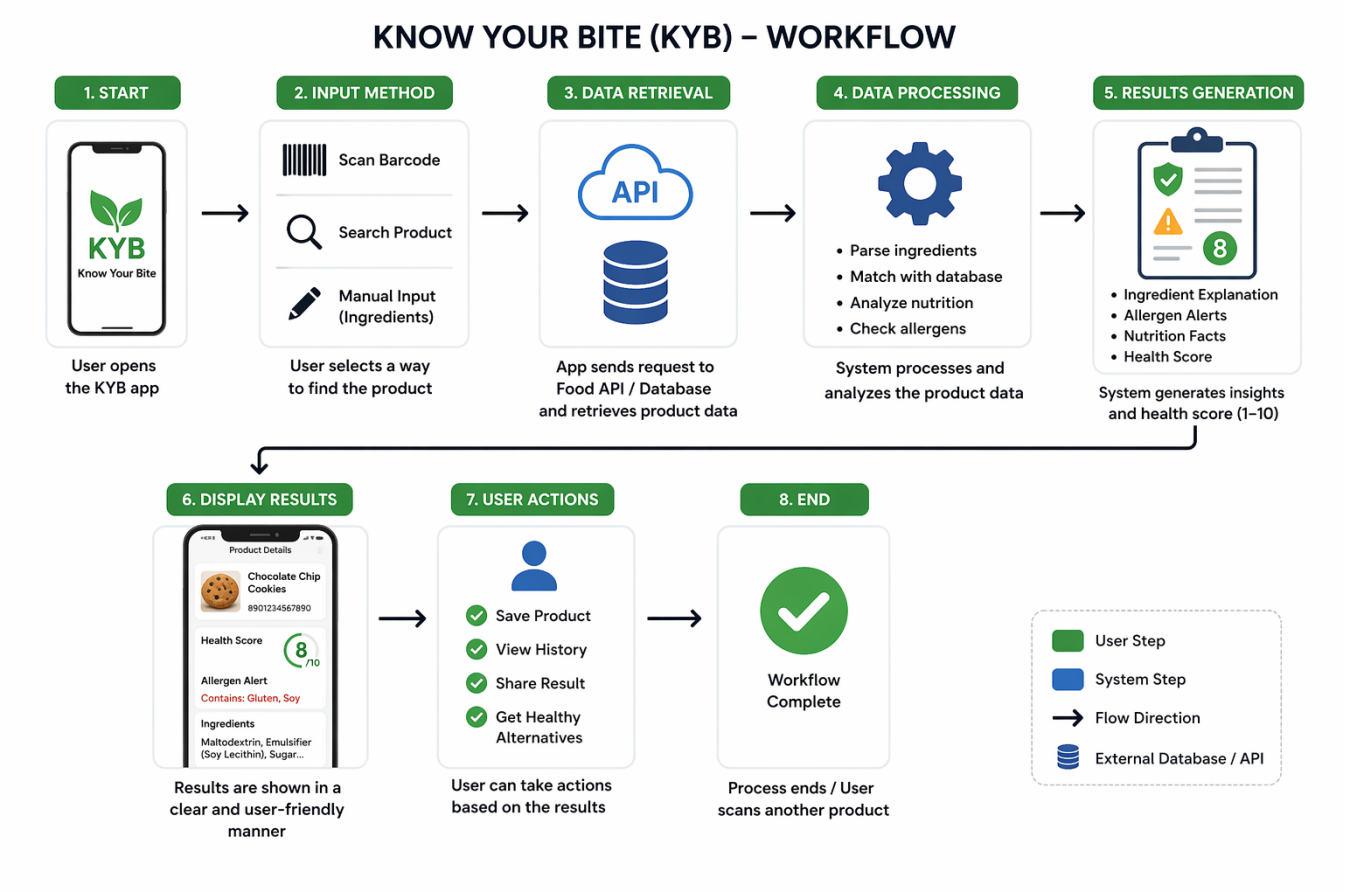

Figure 2 - Workflow of KYB (Know Your Bite)

The **Figure 2 - Workflow of KYB (Know Your Bite)** application illustrates the complete process through which a food product is analyzed and meaningful health insights are delivered to the user. The process begins when the user opens the application, which initializes the system and prepares it to accept input. This marks the starting point of interaction between the user and the system.

In the next stage, the user selects an appropriate input method to analyze a product. The application provides multiple options such as scanning a barcode using the mobile camera, searching for a product by name, or manually entering ingredient details. This flexibility ensures that the application remains user-friendly and accessible in different situations, even when barcode scanning is not possible.

Once the input is provided, the system proceeds to the data retrieval phase. In this step, the application communicates with external food databases through APIs to fetch relevant product information. This includes ingredient lists, nutritional values, and other product-related details. The use of reliable databases ensures that the information provided to the user is accurate and up-to-date.

After retrieving the data, the system performs data processing, which is the core functionality of the application. The ingredients are parsed and matched with an internal database to simplify complex chemical names into understandable terms. At the same time, the system analyzes nutritional values such as sugar, fat, and sodium levels, and checks for the presence of common allergens. This step transforms raw data into meaningful insights.

Following processing, the system generates results in a structured and user-friendly format. It provides simplified ingredient explanations, highlights potential allergens through alerts, displays nutritional information, and assigns a health score ranging from 1 to 10. This score helps users quickly evaluate the overall healthiness of the product without needing deep technical knowledge.

The results are then displayed on a clean and intuitive mobile interface, ensuring that users can easily interpret the information. Visual indicators and clear text enhance readability, making it easier for users to make informed decisions regarding their food choices.

Finally, the user is given options to take further actions based on the results. These include saving the product for future reference, viewing previously scanned items, sharing the results, or exploring healthier alternatives. The workflow concludes after this stage, but it is designed in a way that allows the user to repeat the process seamlessly for another product, ensuring continuous usability of the application.

Chapter 6 : Project Scheduling

The development of the Know Your Bite (KYB) application was carried out using a well-structured and systematic project schedule. The entire project timeline was divided into multiple phases, including planning, design, development, and testing, to ensure efficient utilization of time and resources. Each phase was assigned specific tasks and milestones, allowing for better tracking of progress and timely completion within the academic semester.

During the initial phase, the focus was on requirement analysis and problem identification. This included understanding user needs, defining system objectives, and studying existing solutions. In parallel, dataset preparation was carried out by identifying reliable food databases and structuring the ingredient information required for the application. This phase laid the foundation for the overall system design.

The next phase involved system design and planning. In this stage, the architecture of the application was defined, including the division into presentation, logic, and data layers. UI/UX design was also initiated, where wireframes and screen layouts were prepared to ensure a user-friendly interface. Proper planning at this stage helped reduce complexity during development.

The development phase formed the core part of the project timeline. It included frontend development using Flutter for building responsive user interfaces and backend integration through REST APIs for fetching product data. Features such as barcode scanning, product search, ingredient explanation, allergy detection, and health score calculation were implemented during this stage. Continuous testing was also performed alongside development to identify and fix issues early.

In the final phase, the focus shifted to testing, debugging, and optimization. Functional testing was conducted to ensure that all features were working correctly, while UI refinement improved the overall user experience. Performance optimization and bug fixing were carried out to enhance the reliability and efficiency of the application. Documentation and final report preparation were also completed during this stage.

This structured and phased scheduling approach ensured smooth execution of the project, minimized delays, and helped achieve all objectives within the given academic timeline. The following Gantt Chart visually represents the distribution of tasks and their duration across the project lifecycle, providing a clear overview of the project progress.

Project Implementation Schedule :

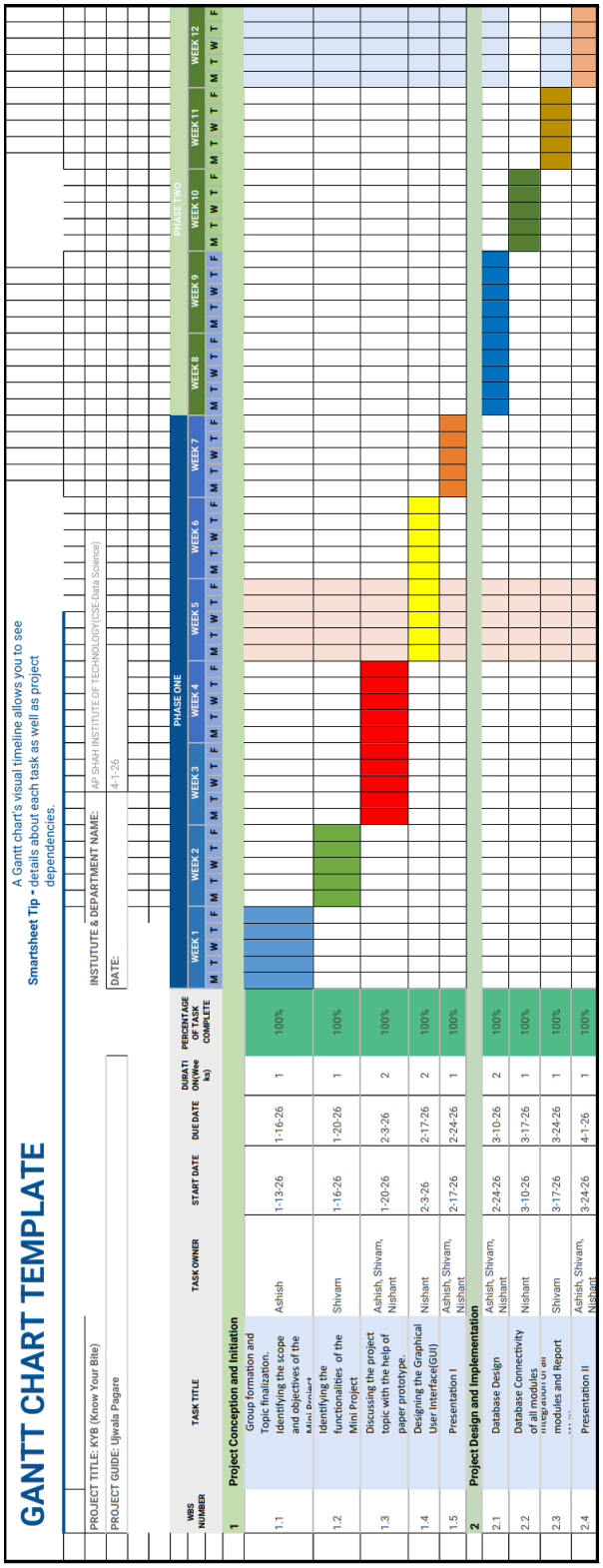

Figure 3 - Gant Chart Diagram.

The Figure 3 – Gantt Chart Diagram illustrates the project schedule of the Know Your Bite (KYB) application across a 12-week timeline, divided into two major phases: Project Conception and Initiation, and Project Design and Implementation. The initial weeks (Week 1–Week 2) focus on topic finalization, scope definition, and identifying system functionalities. During Weeks 3 and 4, activities such as project discussion, prototype planning, and documentation are carried out. The mid-phase (Weeks 5–7) involves designing graphical user interfaces and preparing presentations. In Phase Two (Weeks 8–12), the focus shifts to technical development, including database design, API integration, and connectivity modules, followed by report writing and final presentation. The chart uses color-coded blocks to represent task duration and progress, ensuring clear visualization of task distribution, dependencies, and timely completion within the academic schedule.

Chapter 7 : Results

The system demonstrated effective performance in retrieving and analyzing food product data. Most scanned or searched products were successfully processed with accurate ingredient mapping. The application provided clear health insights, warnings, and personalized recommendations. Users found the interface intuitive and the analysis helpful in decision-making. Overall, the results validate the feasibility and usefulness of the KYB application.

7.1. Scanning Accuracy

Testing was conducted on various packaged goods (biscuits, soft drinks, and snacks). The system successfully retrieved data for **92%** of scanned barcodes using the Open Food Facts API.

7.2. Ingredient Analysis Performance

The matching algorithm used to link API ingredient strings with our internal "Explanation Database" showed high efficiency.

7.3. User Feedback

Initial alpha testing indicated that the **Allergy Safety Alert** was the most valued feature, providing alerts within **<1.5 seconds** of scanning a product.

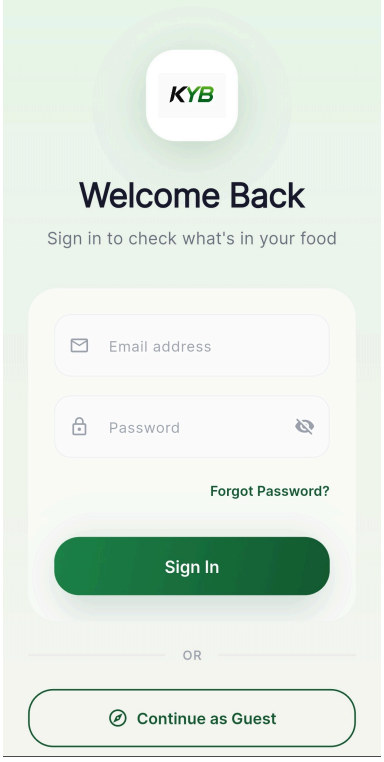

Figure 7.1 - Login Page

This Figure 7.1 - Login Page represents the login interface of the KYB (Know Your Bite) application, where users can securely sign in using their email and password. It also provides a "Forgot Password" option for account recovery and a "Continue as Guest" feature for quick access without registration. The clean and minimal design ensures easy navigation and a smooth user experience.

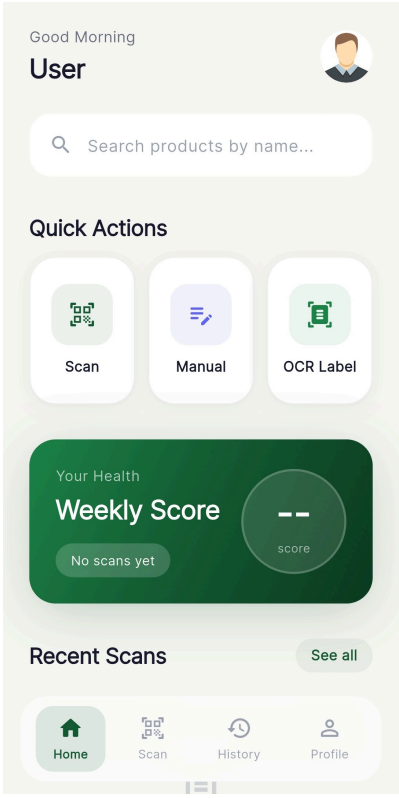

Figure 7.2 - Home Page

This Figure 7.2 - Home Page represents the home page of the KYB (Know Your Bite) application, where users can easily access key features. It includes options like scanning a product, manual input, and OCR label detection for quick analysis. The page also displays a weekly health score and recent scans, helping users track their food choices. The layout is clean and intuitive, ensuring smooth navigation.

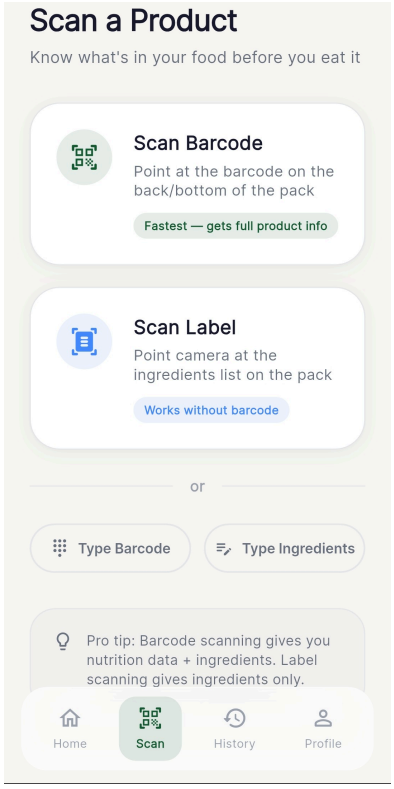

Figure 7.3 - Product Scanning Interface

This Figure 7.3 - Product Scanning Interface represents the product scanning section of the KYB application, where users can analyze food items using different input methods. It provides options to scan a barcode for complete product details or scan the label using OCR when a barcode is not available. Additionally, users can manually enter barcode or ingredient information. The interface is simple and informative, guiding users to choose the most efficient method for accurate food analysis.

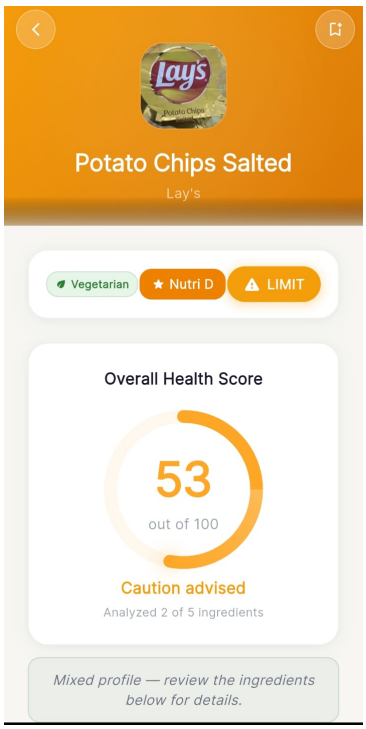

Figure 7.4 - Product Analysis & Health Score Interface

This Figure 7.4 - Product Analysis & Health Score Interface shows the detailed analysis of a scanned product in the KYB application. It displays product information along with tags like vegetarian status and nutrition grade. The main highlight is the health score, which evaluates the product based on its ingredients and nutritional values. The interface clearly indicates whether the product should be consumed with caution, helping users make informed food choices.

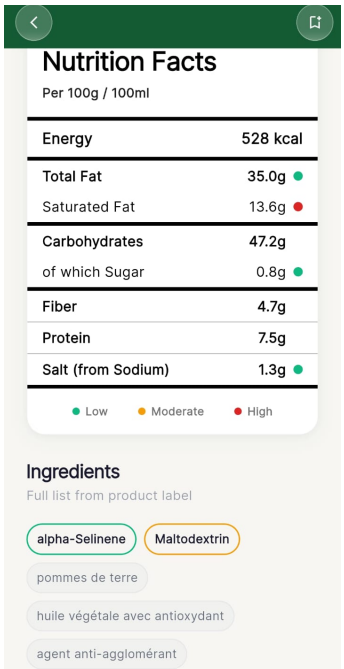

Figure 7.5 - Nutrition Facts

The Nutrition Facts interface in the **(Figure 7.5 - Nutrition Facts)** KYB app acts like a digital label that shows complete food details in one place. It displays values such as calories, proteins, fats, and carbohydrates in a clear and organized way. The UI uses simple colors and icons so users can quickly understand whether the food is healthy or not. It helps users compare items easily and make better diet choices. The design is clean and user-friendly for fast access to important nutrition data.

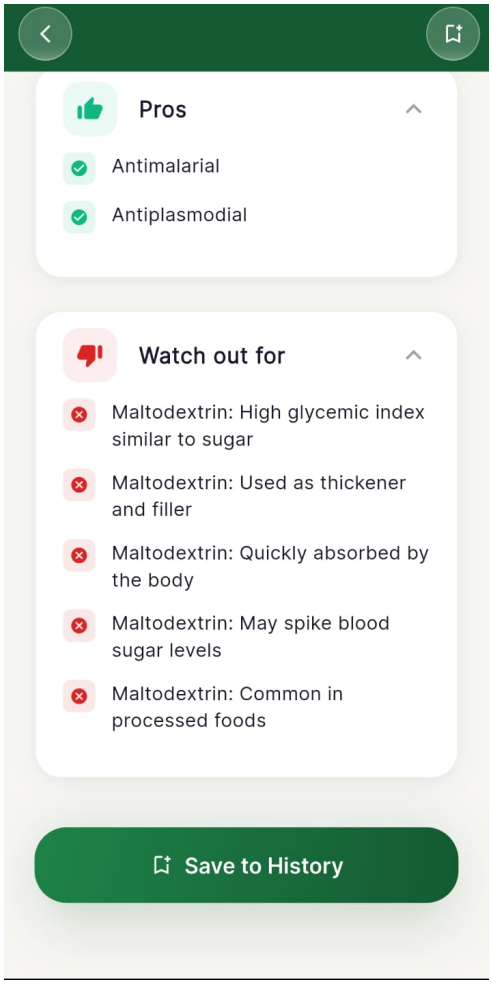

Figure 7.6 - Pros & Cons of Food

This Figure 7.6 - Pros & Cons of Food shows the Pros and Cons section in the KYB app, which highlights the positive and negative aspects of a food item. It clearly lists benefits like high protein or vitamins and drawbacks like high sugar or fat content. The UI uses simple indicators such as colors or icons to differentiate pros and cons. This helps users quickly evaluate the overall health impact of the food. It supports better decision-making by giving a balanced view in a simple format.

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